

Incident handler's journal

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this journal as a way to log the key takeaways about the different cybersecurity tools or concepts you encounter in this course.

Date: October 24, 2026	**Entry:** **#1**
Description	Documenting a cybersecurity incident. This comes under Post-Incident Activity phase of the NIST Incident Response Lifecycle.
Tool(s) used	**None.**
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? An organized group of unethical hackers who are known to target organizations in healthcare and transportation industries. • **What** happened? A ransomware security incident. The company's files are encrypted and they have left a ransom note demanding money in exchange for the decryption key. • **When** did the incident occur? This security incident happened on Tuesday morning, at approximately 9:00 a.m. • **Where** did the incident happen? The incident happened at a small U.S. health care

	clinic specializing in delivering primary-care services. • **Why** did the incident happen? The cause of the security incident was a phishing email that contained a malicious attachment, which led to all the files being encrypted. They left a ransom note demanding money in exchange for the decryption key.
Additional notes	1. How could the health care company prevent an incident like this from occurring again? 2. Should the company pay the ransom to retrieve the decryption key?

Date: October 26, 2026	**Entry:** #2
Description	Analyzing packet capture file. This comes under Detection and Analysis phase of the NIST Incident Response Lifecycle.
Tool(s) used	For this activity I am using Wireshark tool. This tool has a graphical user interface which can help to inspect packet data and apply filters to sort through packet information efficiently. This can help in detecting and investigating malicious activity.
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? N/A • **What** happened? N/A • **When** did the incident occur? N/A

	• **Where** did the incident happen? N/A • **Why** did the incident happen? N/A
Additional notes	I have never used the Wireshark tool before, but this GUI is user friendly. By inspecting data packets, we can find and filter a lot of information, like source IP address and port, destination IP address and port, protocols, MAC address, etc., which can be helpful for a security analyst.

Date: October 27, 2026	**Entry:** #3
Description	Capturing and analyzing live network traffic. This comes under Detection and Analysis phase of the NIST Incident Response Lifecycle.
Tool(s) used	For this activity, I am using tcpdump tool which is a command line interface tool.
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? N/A • **What** happened? N/A • **When** did the incident occur? N/A • **Where** did the incident happen? N/A • **Why** did the incident happen?

	N/A
Additional notes	Using tcpdump can be quite challenging since it has a command-line interface, which is new to me, but this tool can give more control to the user. I captured live network traffic and saved the capture file and then I analyzed this captured file for information regarding network traffic.

Date: October 29, 2026	**Entry:** #4
Description	Investigating file hash of a suspicious file. This comes under Detection and Analysis phase of the NIST Incident Response Lifecycle.
Tool(s) used	VirusTotal
The 5 W's	Capture the 5 W's of an incident. ● **Who** caused the incident? A malicious actor ● **What** happened? A malicious payload was executed on the computer. ● **When** did the incident occur? From 1:11 p.m. to 1:20 p.m., a series of events took place - **1:11 p.m.:** An employee receives an email containing a file attachment. - **1:13 p.m.:** The employee successfully downloads and opens the file. - **1:15 p.m.:** Multiple unauthorized executable files are created on the employee's computer. - **1:20 p.m.:** An intrusion detection system detects the executable files and sends out an alert to

	the SOC. • **Where** did the incident happen? The incident happened at a financial services company • **Why** did the incident happen? An employee downloaded and opened an attachment that came through the email. After opening the file, a malicious payload was executed on the computer, and the IDS detected this and sent an alert to the SOC analyst.
Additional notes	After confirming the file is malicious, we followed the incident response plan by following the playbook which the organization has created for this kind of scenario. By following the playbook’s step-by-step actions, we can minimize the impact of an incident and reduce the incident response time. How can we prevent this in the future? Would implementing employee training to identify phishing emails stop this kind of phishing-related security event?

Date: November 1, 2026	**Entry:** #5
Description	Reviewing an incident final report. This comes under Post-Incident Activity phase of the NIST Incident Response Lifecycle.
Tool(s) used	None.
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? Due to a vulnerability in the e-commerce web

	application, this incident took place. • **What** happened? On December 22, 2022, an employee received an email from an external email address claiming they have successfully stolen customer data and in exchange for not releasing the data to public forums they requested a $25,000 cryptocurrency payment. • **When** did the incident occur? December 28, 2022, at 7:20 p.m. PT • **Where** did the incident happen? A mid-sized retail company • **Why** did the incident happen? The root cause of the incident was identified as a vulnerability in the e-commerce web application. This vulnerability allowed the attacker to perform a forced browsing attack and access customer transaction data by modifying the order number included in the URL string of a purchase confirmation page. This vulnerability allowed the attacker to access customer purchase confirmation pages, exposing customer data, which the attacker then collected and exfiltrated.
Additional notes	I learned how to review and got an idea of how to write a final report from this activity. After reviewing this final report of an incident, I recommended some security measures to prevent any this kind of security event which I learned from this course like vulnerability scanning, conducting penetration test, implementing allowlisting, and following least privilege.

Date:	**Entry:**

November 4, 2026	#6
Description	Exploring signatures and logs using Suricata. This comes under Detection and Analysis phase of the NIST Incident Response Lifecycle.
Tool(s) used	Suricata
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? N/A • **What** happened? N/A • **When** did the incident occur? N/A • **Where** did the incident happen? N/A • **Why** did the incident happen? N/A
Additional notes	In this activity I learned how Suricata works as an IDS. Suricata should be configured according to our requirements using a custom signature file. Suricata will then apply the rules to the packets that pass through the interface, which determines whether each packet should generate an alert. Then it will create an output log file which we can filter and analyze.

Date: November 6, 2026	**Entry:** #7
Description	Query for events with Splunk. This comes under Detection and Analysis phase of the NIST Incident Response Lifecycle.
Tool(s) used	Splunk
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? N/A • **What** happened? N/A • **When** did the incident occur? N/A • **Where** did the incident happen? N/A • **Why** did the incident happen? N/A
Additional notes	Splunk has its own querying language called Search Processing Language (SPL). An SPL search can contain many different commands and arguments.

Date: November 7, 2026	**Entry:** #8
Description	Query for events with Google SecOps. This comes under Detection and Analysis phase of the NIST Incident Response Lifecycle.

Tool(s) used	Google SecOps(Chronicle)
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? N/A • **What** happened? N/A • **When** did the incident occur? N/A • **Where** did the incident happen? N/A • **Why** did the incident happen? N/A
Additional notes	In Google SecOps (Chronicle), we can search for events using the Search field. There are two types of searches we can perform to find events in Google SecOps, a Unified Data Model (UDM) Search or a Raw Log Search. Through a UDM Search, Google SecOps searches security data that has been ingested, parsed, and normalized. A UDM Search retrieves search results faster than a Raw Log Search because it searches through indexed and structured data that's normalized. Raw Log Search will search through the raw, unparsed logs. Because it is searching through raw logs, it takes longer than a structured search.

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If you want to add more journal entries, please copy one of the tables above and paste it into the template to use for future entries.

Reflections/Notes:

1. **Were there any specific activities that were challenging for you? Why or why not?**

I really found the activity using tcpdump challenging. I am new to using the command line, and learning the syntax for a tool like tcpdump was a

big learning curve. What I learned from this was to carefully read the instructions and work through the process slowly.

2. **Has your understanding of incident detection and response changed after taking this course?**

After taking this course, my understanding of incident detection and response has definitely evolved. At the beginning of the course, I had some basic understanding of what detection and response entailed, but I didn't fully understand the complexity involved. As I progressed through the course, I learned about the lifecycle of an incident; the importance of plans, processes, and people; and tools used. Overall, I feel that my understanding has changed, and I am equipped with more knowledge and understanding about incident detection and response.

3. **Was there a specific tool or concept that you enjoyed the most? Why?**

I liked how the SIEM tool worked, like how it collects all the log data from multiple sources and centralizes it on the dashboard. I also like how it can display it as a chart. Since we work on it daily as a security analyst, I focused on it more to learn this tool.